Unit 4

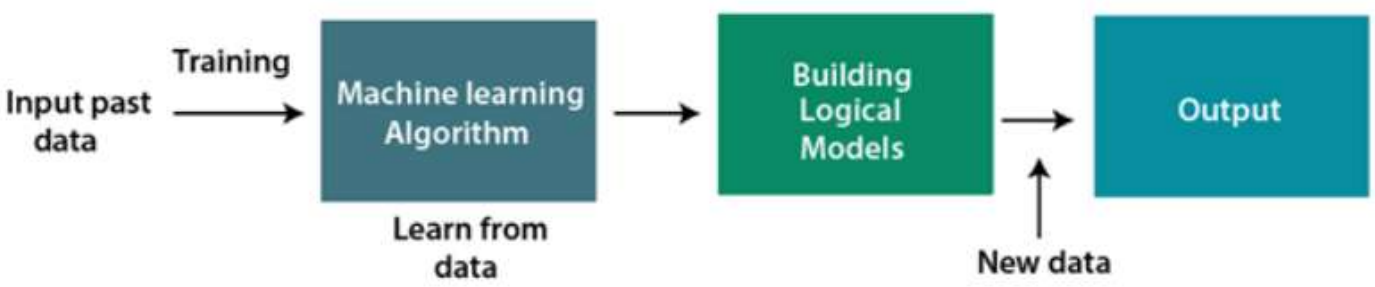
Machine learning

Machine Learning

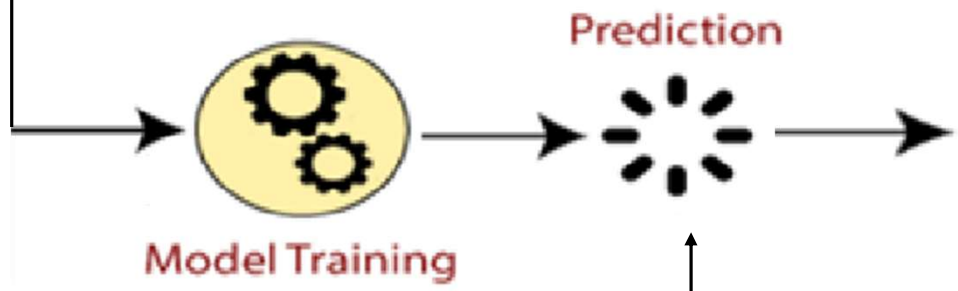
- Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.

How does Machine Learning work

- A Machine Learning system **learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it**. The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.

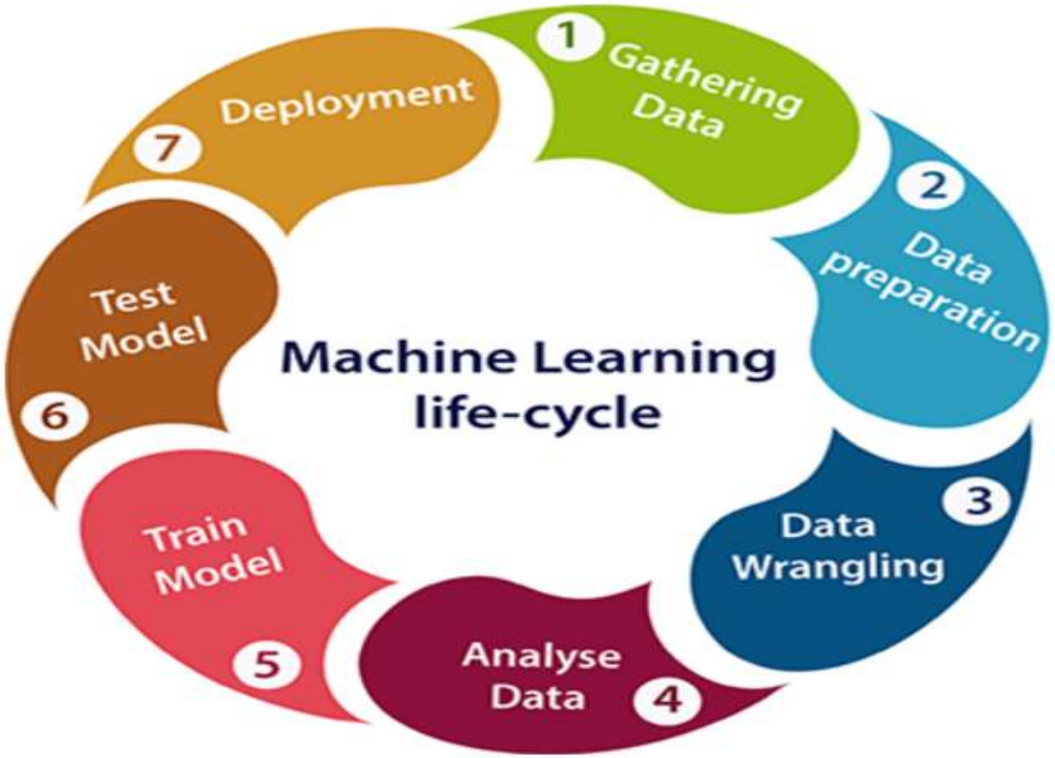


	Data Point	x	y
	1	1	2
	2	3	4
	3	5	6
	4	7	8
80%	5	9	10
	6	11	12
	7	13	14
	8	15	16
	Data Point	**X'**	**Y'**
	9	17	18
20%	10	19	20
	11	21	22
	12	23	24



y'
18.1
20.4
22.0
24.2

Machine learning Life cycle



1. **Gathering Data:**

- Identify various data sources
- Collect data
- Integrate the data obtained from different sources

2. **Data preparation:**

- Data exploration
- Data pre-processing

3. **Data Wrangling**

- Missing Values
- Duplicate data
- Invalid data
- Noise

4. **Data Analysis**

- Selection of analytical techniques
- Building models
- Review the result

5. **Train Model**

Now the next step is to train the model, in this step we train our model to improve its performance for better outcome of the problem.

6. **Test Model**

Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.

7. **Deployment**

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

Classification of Machine Learning

At a broad level, machine learning can be classified into three types:

1.Supervised learning

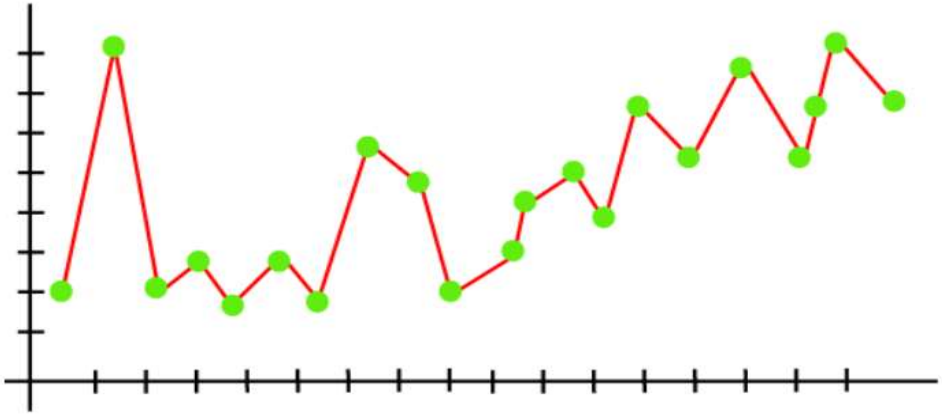
2.Unsupervised learning

3.Reinforcement learning

Overfitting

- Overfitting occurs when our machine learning model tries to cover all the data points or more than the required data points present in the given dataset. Because of this, the model starts caching noise and inaccurate values present in the dataset, and all these factors reduce the efficiency and accuracy of the model.
- An overfitted model learns the training data too well (low bias) but becomes overly sensitive to small changes, making its predictions unreliable on new data (high variance).The overfitted model has **low bias** and **high variance.**
- The chances of occurrence of overfitting increase as much we provide training to our model. It means the more we train our model, the more chances of occurring the overfitted model.
- Overfitting is the main problem that occurs in supervised learning.

Example: The concept of the overfitting can be understood by the below graph of the linear regression output:



Scenario	Training Features	Test Example	Outcome	Explanation
Overfitting	- **Sphere:** Checks if the object is spherical. - **Play:** Determines if the object is used for playing. - **Eat:** Verifies that the object is inedible. - **radius = 5 cm:** Strict size condition.	**Basketball** (spherical, playable, inedible, but with a radius of 15 cm)	Predicted as not a ball	The model is given many detailed features. It checks every attribute and finds that although most conditions are met, the strict radius condition (5 cm) fails when tested with a basketball (15 cm). This rigidity causes misclassification.

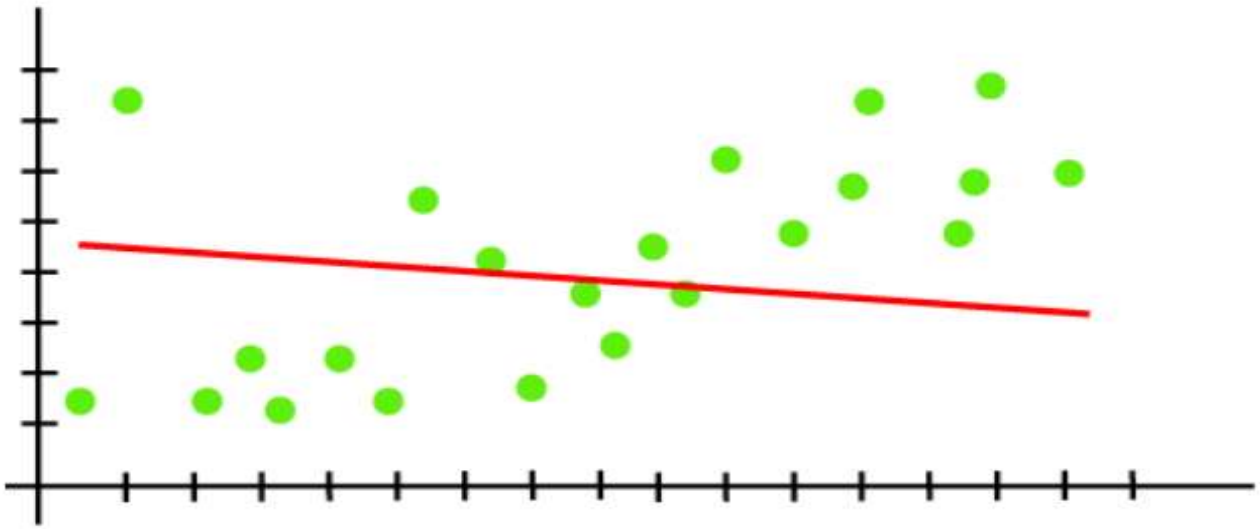
Underfitting

Underfitting occurs when our machine learning model is not able to capture the underlying trend of the data. To avoid the overfitting in the model, the fed of training data can be stopped at an early stage, due to which the model may not learn enough from the training data. As a result, it may fail to find the best fit of the dominant trend in the data.

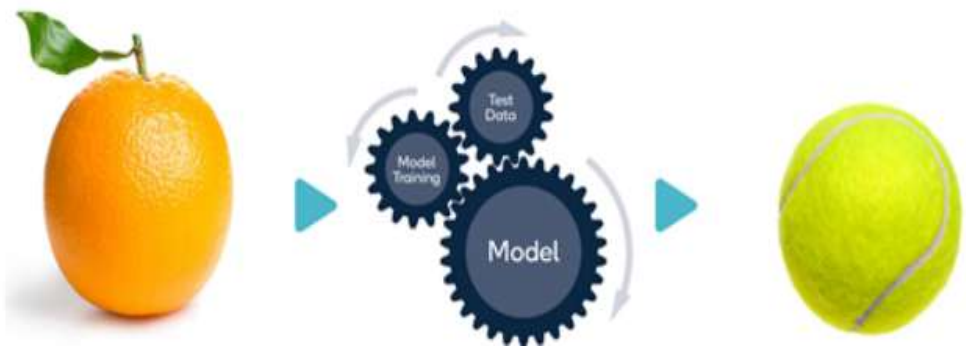
In the case of underfitting, the model is not able to learn enough from the training data, and hence it reduces the accuracy and produces unreliable predictions.

An underfitted model has high bias and low variance.

Example: We can understand the underfitting using below output of the linear regression model:

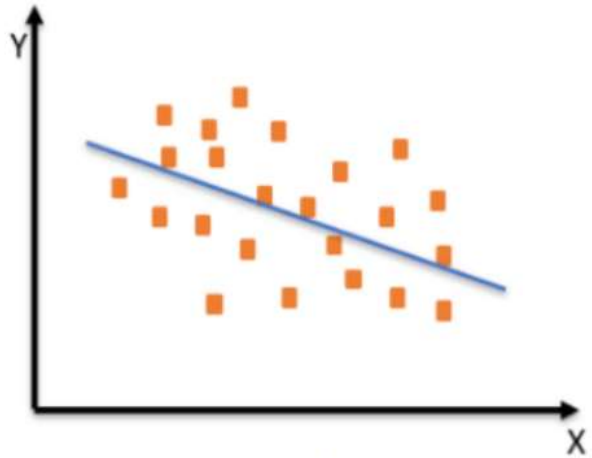


Scenario	Training Features	Test Example	Outcome	Explanation
Underfitting	**Circle:** Checks if the object is circular.	**Orange** (a circular object)	Predicted as a ball	With only the “Circle" feature, the model oversimplifies the concept of a ball. It classifies any circular object as a ball—even an orange—because it hasn't learned enough distinguishing details.



Good Fit Model

A good fit model is a balanced model, which is not suffering from underfitting and overfitting. This is a perfect model which gives good accuracy score during training and equally performs well during testing.



Good Fit

Scenario	Training Features	Test Example	Outcome	Explanation
Good Fit (Balanced Model)	- **Sphere:** Confirms the object is spherical. - **Play:** Indicates it's used for playing. - **Eat:** Ensures it's inedible. - **Radius Range:** Accepts a flexible radius (e.g., between 5 cm and 20 cm).	Basketball (spherical, playable, inedible, radius = 15 cm)	Correctly predicted as a ball	The model uses a balanced set of features, including a flexible condition on radius rather than a rigid value. This allows it to capture the essential attributes of a ball without over-relying on a single detail, thereby generalizing well to different types and sizes of balls.

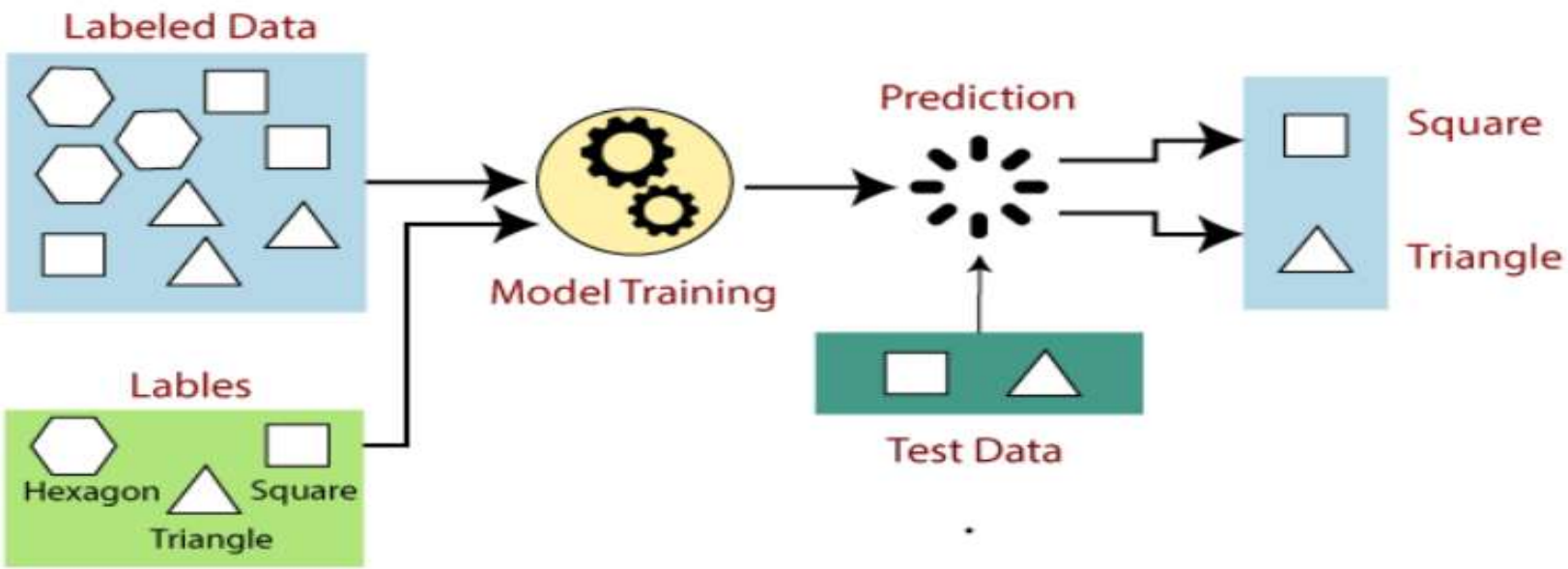
Supervised Machine Learning

- Supervised learning is the types of machine learning in which machines are trained using well **"labelled"** training data, and on basis of that data, machines predict the output. The labelled data means some input data is already tagged with the correct output.

- Supervised learning is a process of providing input data as well as correct output data to the machine learning model. The aim of a supervised learning algorithm is to **find a mapping function to map the input variable(x) with the output variable(y)**.

- In the real-world, supervised learning can be used for **Risk Assessment, Image classification, Fraud Detection, spam filtering**, etc.

Working of Supervised Learning



Types of supervised Machine learning Algorithms:

1. **Regression:-** In machine learning, **a regression problem is the problem of predicting the value of a numeric variable based on observed values of the variable**. The value of the output variable may be a number, such as an integer or a floating-point value. These are often quantities, such as amounts and sizes. The input variables may be discrete or real-valued.

   Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc. Regression algorithms which come under supervised learning:
   - Linear Regression
   - Regression Trees
   - Non-Linear Regression
   - Bayesian Linear Regression
   - Polynomial Regression

2. Classification:- In machine learning, **classification is the problem of identifying to which of a set of categories a new observation belongs**, on the basis of a training set of data containing observations (or instances) whose category membership is known. Classification algorithms are used when the output variable is category base , which means there are two classes such as Yes-No, Male-Female, True-false, etc.

•Random Forest
•Decision Trees
•Logistic Regression
•Support vector Machines

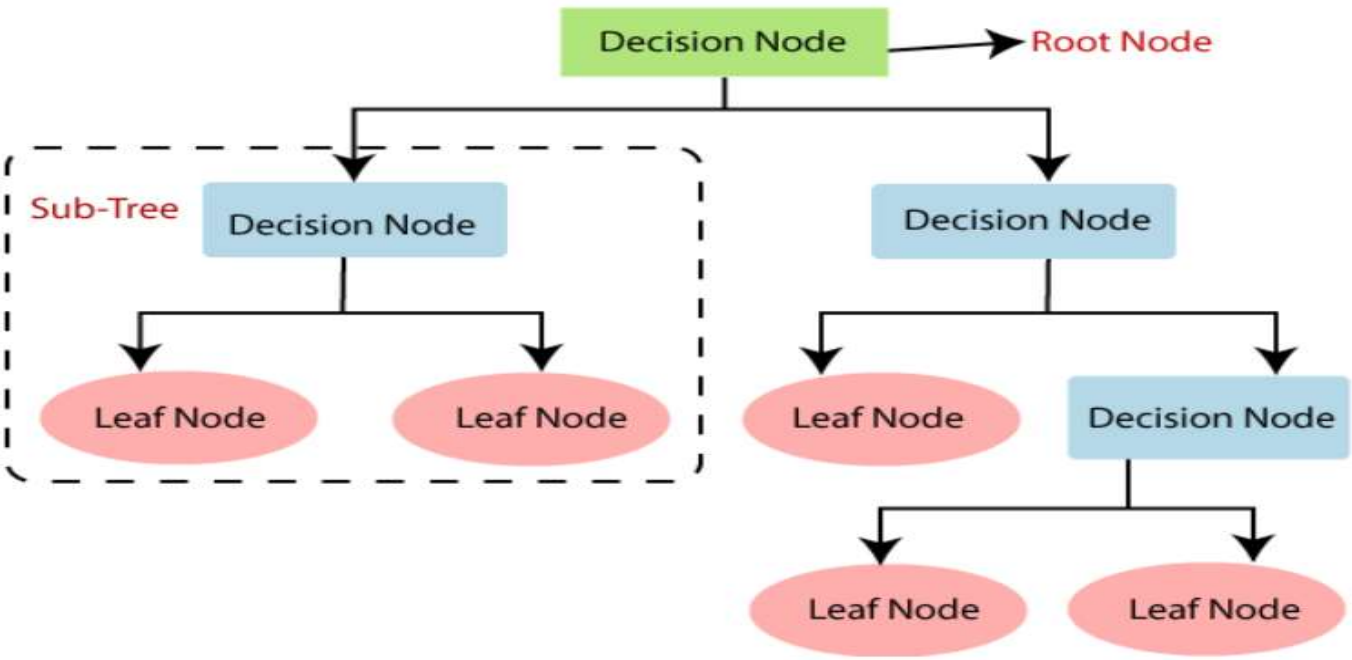
Remarks

- A **classification problem** requires that examples be classified into one of two or more classes.
- A **classification task** can have **real-valued** or **discrete** input variables.
- A problem with **two classes** is often called a **two-class** or **binary classification** problem.
- A problem with **more than two classes** is often called a **multi-class classification** problem.
  Example: Classifying a disease as **Diabetes, Hypertension, or Heart Disease.**
- A problem where an example is assigned **multiple classes** is called a **multi-label classification** problem.

Movie	Multi-Class Classification (Single Label)	Multi-Label Classification (Multiple Labels)
Frozen	Animation	Animation, Fantasy, Musical
Avengers: Endgame	Sci-Fi	Action, Sci-Fi, Adventure
Inception	Thriller	Sci-Fi, Thriller, Mystery

Decision Tree Classification Algorithm

Decision Trees are a type of Supervised Machine Learning (that is you explain what the input is and what the corresponding output is in the training data) where the data is continuously split according to a certain parameter. The tree can be explained by two entities, namely decision nodes and leaves. The leaves are the decisions or the final outcomes. And the decision nodes are where the data is split.



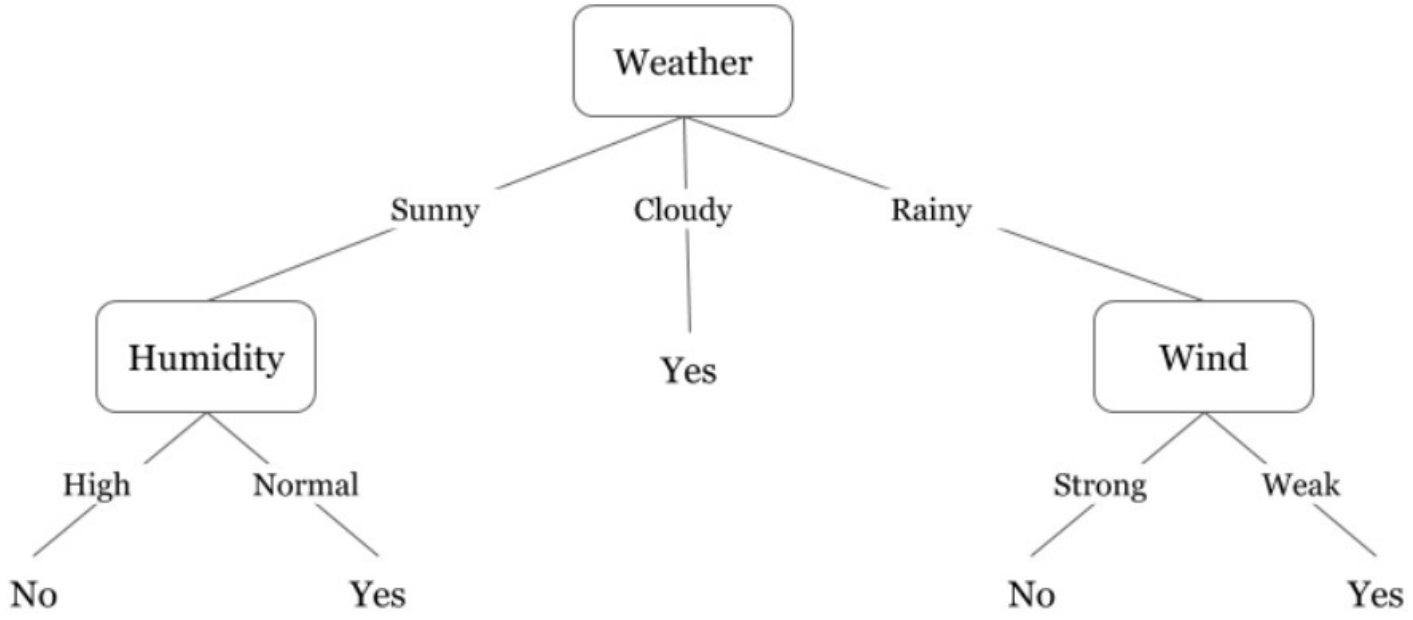
Decision Tree Terminologies

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.

Example:- Let's assume we want to play badminton on a particular day — say Saturday — how will you decide whether to play or not. Let's say you go out and check if it's hot or cold, check the speed of the wind and humidity, how the weather is, i.e. is it sunny, cloudy, or rainy. You take all these factors into account to decide if you want to play or not.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

A decision tree for the concept Play Badminton

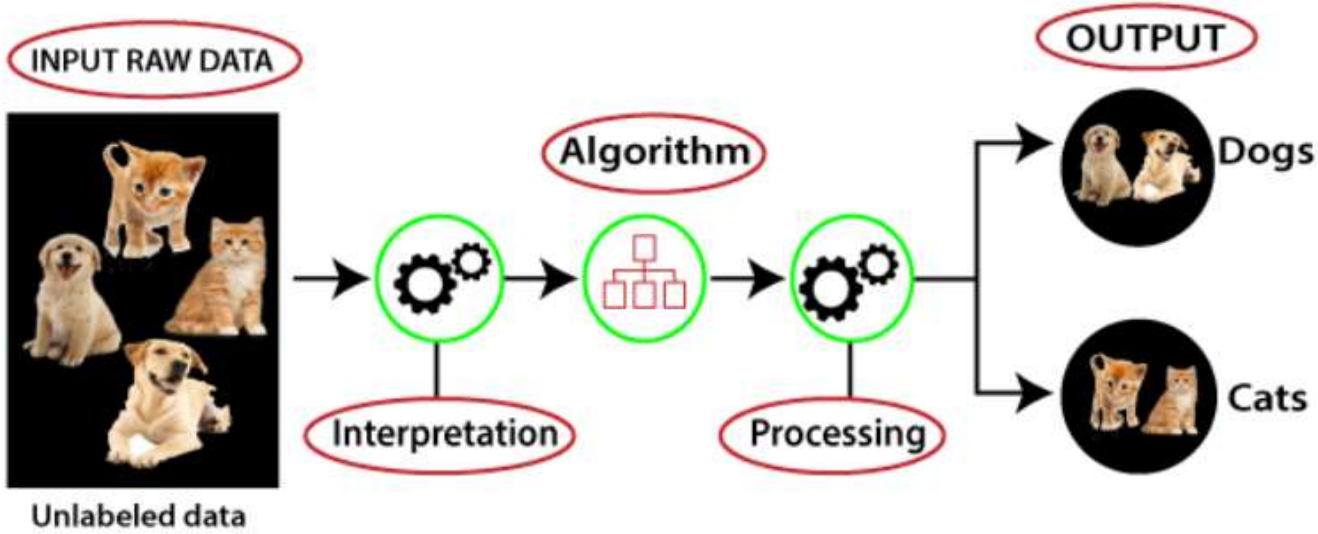


Unsupervised Learning

- Unsupervised learning is a type of machine learning in which models are trained using unlabeled dataset and are allowed to act on that data without any supervision.

- The goal of unsupervised learning is to find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format.

Working of Unsupervised Learning



Use Unsupervised Learning

- Unsupervised learning is helpful for finding useful insights from the data.

- Unsupervised learning is much similar as a human learns to think by their own experiences, which makes it closer to the real AI.

- Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.

- In real-world, we do not always have input data with the corresponding output so to solve such cases, we need unsupervised learning.

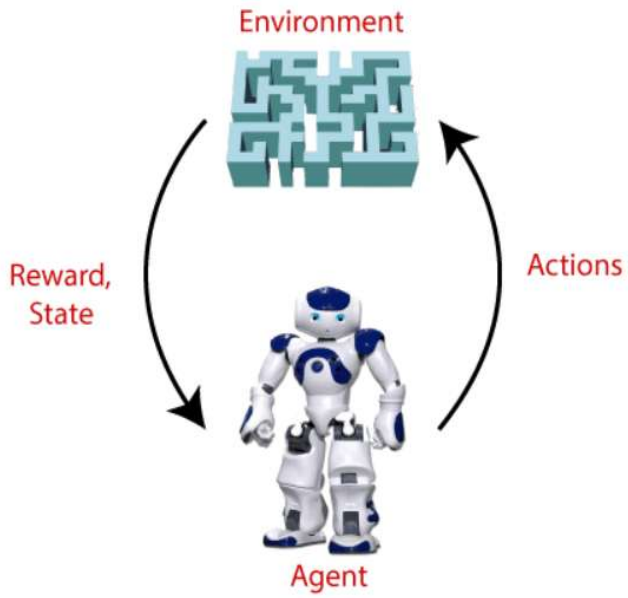
Reinforcement learning

Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance.

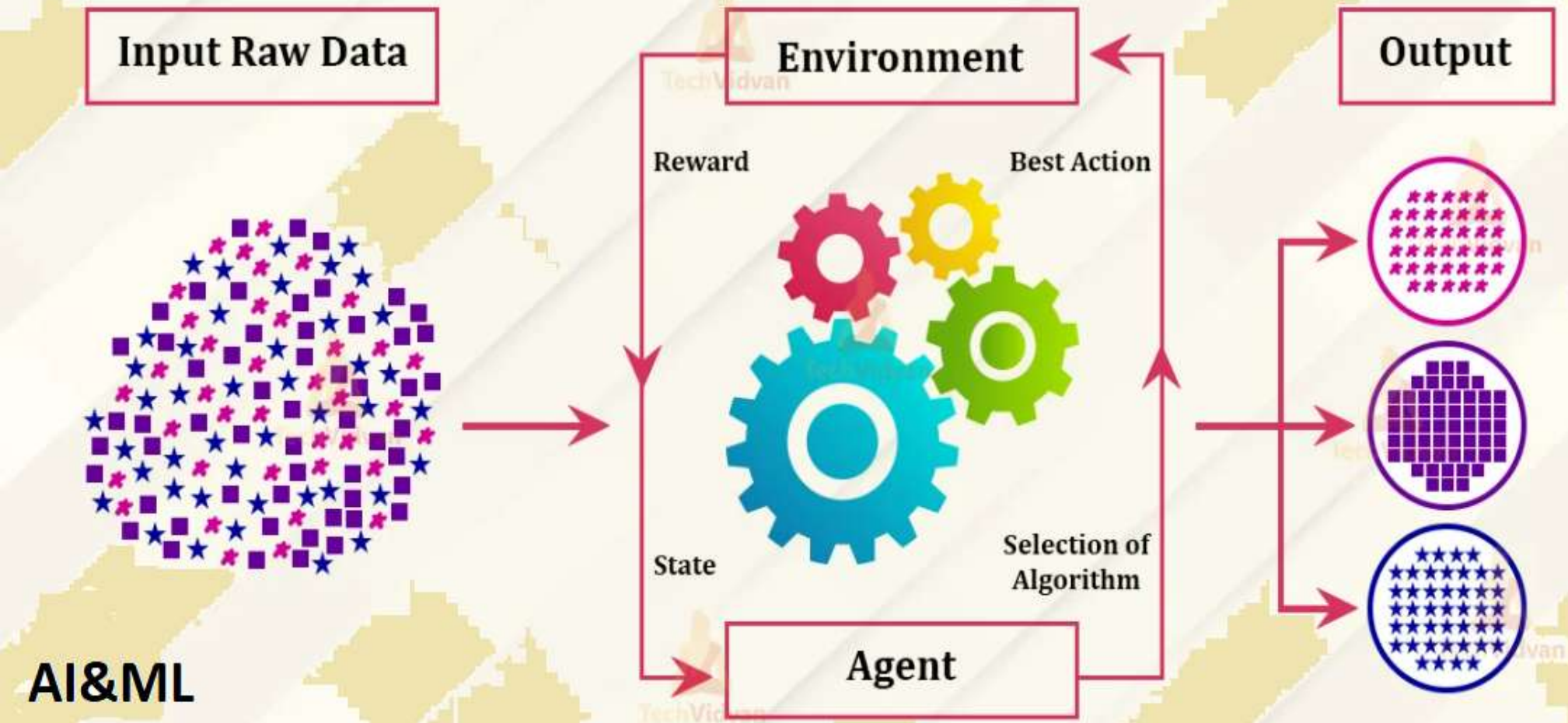
Reinforcement learning is a type of machine learning method where an intelligent agent (computer program) interacts with the environment and learns to act within that.

The agent interacts with the environment by taking actions, receiving rewards, and observing the resulting states. The agent then uses this information to update its policy and make better decisions in the future.

Here is an example of a reinforcement learning problem: training an agent to play a game of chess. The agent takes actions by moving pieces on the board, and the environment responds by updating the state of the board and providing a reward signal based on the outcome of the move. The agent's goal is to learn a policy that maximizes its chances of winning the game. The agent uses this feedback to update its policy, so that it is more likely to make winning moves in the future.



Reinforcement Learning in ML



AI&ML

Performance Metrics in Machine Learning

To evaluate the performance or quality of the model, different metrics are used, and these metrics are known as performance metrics or evaluation metrics. These performance metrics help us understand how well our model has performed for the given data. In this way, we can improve the model's performance by tuning the hyper-parameters. Each ML model aims to generalize well on unseen/new data, and performance metrics help determine how well the model generalizes on the new dataset.

Performance Metrics for Classification

- **Accuracy**
- **Confusion Matrix**
- **Precision**
- **Recall**
- **F-Score**

Confusion Matrix in Machine Learning

Confusion Matrix helps us to display the performance of a model or how a model has made its prediction in Machine Learning.

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

Confusion Matrix for Binary Classification

This matrix consists of 4 main elements that show different metrics to count a number of correct and incorrect predictions. Each element has two words either as follows:

- True or False
- Positive or Negative

There are four metrics combinations in the confusion matrix, which are as follows:

•**True Positive:** This combination tells us how many times a model correctly classifies a positive sample as Positive?

•**False Negative:** This combination tells us how many times a model incorrectly classifies a positive sample as Negative?

•**False Positive:** This combination tells us how many times a model incorrectly classifies a negative sample as Positive?

•**True Negative:** This combination tells us how many times a model correctly classifies a negative sample as Negative?

True Positive Predicted Positive Actual Positive	**False Negative** Predicted Negative Actual Positive
False Positive Predicted Positive Actual Negative	**True Negative** Predicted Negative Actual Negative

Precision(P):.*Precision* is a measure of how many of the positive predictions made are correct (true positives).
The fraction of true positives (TP, correct predictions) from the total amount of relevant results, i.e., the sum of TP and false positives (FP). For multi-class classification problems, P is averaged among the classes. The following is the formula for precision. **Precision is the ratio between the True Positives and all the Positives**

$$P=TP/(TP+FP)$$

Recall(R):*Recall* is a measure of how many of the positive cases the classifier correctly predicted, over all the positive cases in the data. (**The recall is the measure of our model correctly identifying True Positives**)
The fraction of TP from the total amount of TP and false negatives (FN). For multi-class classification problems, R gets averaged among all the classes. The following is the formula for recall.

$$R=TP/(TP+FN)$$

F1 score(F1):
The harmonic mean of precision and recall. For multi-class classification problems, F1 gets averaged among all the classes. The following is the formula for F1 score.

$$F1=2 * (Precision * Recall)/(Precision + Recall)$$

Accuracy:
Accuracy is a metric that generally describes how the model performs across all classes.

$$(TP+TN)/Dataset\ Size$$

Metric Name	Metric Forumla	Code	When to use
Accuracy	$\textbf{Accuracy} = \frac{tp + tn}{tp + tn + fp + fn}$	`tf.keras.metrics.Accuracy()` or `sklearn.metrics.accuracy_score()`	Default metric for classification problems. Not the best for imbalanced classes.
Precision	$\textbf{Precision} = \frac{tp}{tp + fp}$	`tf.keras.metrics.Precision()` or `sklearn.metrics.precision_score()`	Higher precision leads to less false positives.
Recall	$\textbf{Recall} = \frac{tp}{tp + fn}$	`tf.keras.metrics.Recall()` or `sklearn.metrics.recall_score()`	Higher recall leads to less false negatives.
F1-score	$\textbf{F1-score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	`sklearn.metrics.f1_score()`	Combination of precision and recall, usually a good overall metric for a classification model.
Confusion matrix	NA	`Custom function` or `sklearn.metrics.confusion_matrix()`	When comparing predictions to truth labels to see where model gets confused. Can be hard to use with large numbers of classes.

For example:-Using the cancer prediction example, a confusion matrix for 100 patients might look something like this:

		Actual (True) Values	
		Cancer	No Cancer
Predicted Values	Cancer	45	18
	No Cancer	12	25

- **Accuracy:-**(TP+TN)/Dataset Size

  (45+25)/100=0.7=70%

- **Precision:-** TP/(TP+FP)

  45/(45+18)=45/63=0.714=71.4%

- **Recall :-** R=TP/(TP+FN)

  45/(45+12)=45/57=0.789=78.9%.

- **F1-Score:-** 2 * (Precision * Recall)/(Precision + Recall)

  2*(0.01*1.0)/(0.01+1.0)=~0.02

Confusion Matrix For binary classification

A 2X2 Confusion matrix is shown below for the image recognition having a Dog image or Not Dog image.

		Actual	
		Dog	Not Dog
Predicted	Dog	True Positive (TP)	False Positive (FP)
	Not Dog	False Negative (FN)	True Negative (TN)

- **True Positive (TP):** It is the total counts having both predicted and actual values are Dog.
- **True Negative (TN):** It is the total counts having both predicted and actual values are Not Dog.
- **False Positive (FP):** It is the total counts having prediction is Dog while actually Not Dog.
- **False Negative (FN):** It is the total counts having prediction is Not Dog while actually, it is Dog.

Example for binary classification problems

- Actual Dog Counts = 6
- Actual Not Dog Counts = 4
- True Positive Counts = 5
- False Positive Counts = 1
- True Negative Counts = 3
- False Negative Counts = 1

Index	1	2	3	4	5	6	7	8	9	10
Actual	Dog	Dog	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Not Dog	Not Dog
Predicted	Dog	Not Dog	Dog	Not Dog	Dog	Dog	Dog	Dog	Not Dog	Not Dog
Result	TP	FN	TP	TN	TP	FP	TP	TP	TN	TN

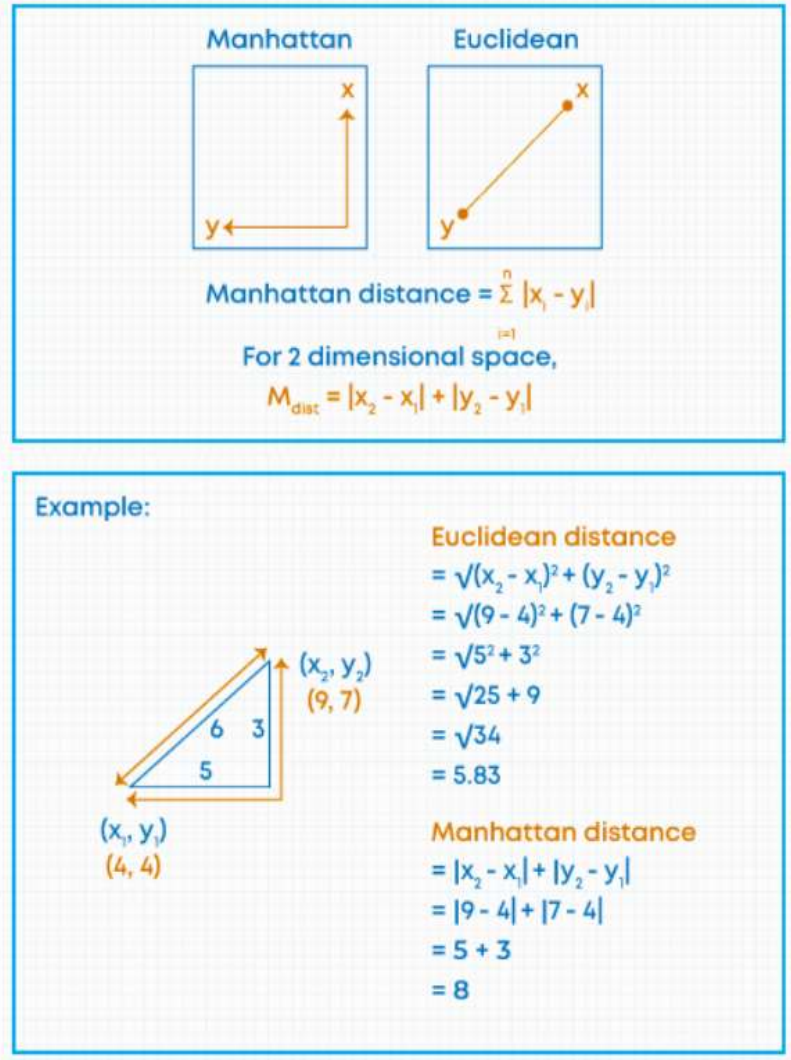
Metrics based on Confusion Matrix Data

1. Accuracy= (5+3)/(5+3+1+1) = 8/10 = 0.8
2. Precision=5/(5+1) =5/6 = 0.8333
3.Recall = 5/(5+1) =5/6 = 0.8333
4. F1-Score: = (2* 0.8333* 0.8333)/(0.8333+ 0.8333) = 0.8333

		Actual	
		Dog	Not Dog
Predicted	Dog	True Positive (TP =5)	False Positive (FP=1)
	Not Dog	False Negative (FN =1)	True Negative (TN=3)

1. Manhattan Distance Calculation:-

- It is the sum of the absolute differences between points across all the dimensions.
- It calculates the distance between real vectors.
- The Manhattan Distance between two points is calculated using a simple formula.



2. Euclidean:-

It is one of the most common distance metrics which is very often used in machine learning algorithms that calculates the distance between two real-valued vectors. It is the shortest distance between two points.

3. Cosine Similarity:- Cosine similarity is a metric used to measure the similarity of two vectors.

$$s(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{\sum_{i=0}^{n-1} x_i y_i}{\sqrt{\sum_{i=0}^{n-1} (x_i)^2} \times \sqrt{\sum_{i=0}^{n-1} (y_i)^2}}$$

- **x . y** = product (dot) of the vectors 'x' and 'y'.
- **||x||** and **||y||** = length of the two vectors 'x' and 'y' or are the magnitudes **(Euclidean norms)** of X and Y.
- **||x|| * ||y||** = cross product of the two vectors 'x' and 'y'.

Example :
Consider an example to find the similarity between two vectors – **'x'** and **'y'**, using Cosine Similarity.
The 'x' vector has values, **x = { 3, 2, 0, 5 }**
The 'y' vector has values, **y = { 1, 0, 0, 0 }**

The formula for calculating the cosine similarity is : **Cos(x, y) = x . y / ||x|| * ||y||**

```
x . y = 3*1 + 2*0 + 0*0 + 5*0 = 3

||x|| = √ (3)^2 + (2)^2 + (0)^2 + (5)^2 = 6.16
||y|| = √ (1)^2 + (0)^2 + (0)^2 + (0)^2 = 1
∴ Cos(x, y) = 3 / (6.16 * 1) = 0.49
```

K-Nearest Neighbor(KNN) Algorithm

K-Nearest Neighbor is one of the simplest Machine Learning algorithms based on Supervised Learning technique. It is used for classification (and regression). It classifies data points based on the majority class of their nearest neighbors.

KNN working:

1.Choose the number of neighbors (K): This is a **hyperparameter** that determines how many nearest points will be considered.
2.Measure the distance: Calculate the distance between the new data point and all existing data points using a distance metric (e.g., Euclidean distance).
3.Identify the K nearest neighbors: Select the closest K points from the training data.
4.Vote for the majority class: The new data point is classified into the class that appears most frequently among the K nearest neighbors.

Importance of k in KNN

- **Noise Reduction & Stability**
  k=1 is sensitive to outliers, while a **optimal** k smooths classification and reduces noise.
- **Balancing Overfitting & Generalization**
  Small k may **overfit**, while a **large** k may **underfit**; choosing an **optimal** k ensures **better accuracy**.

Working of KNN Algorithm

K-nearest neighbors (KNN) algorithm uses ‘feature similarity’ to predict the values of new data points which further means that the new data point will be assigned a value based on how closely it matches the points in the training set.

following steps −

Step 1 − For implementing any algorithm, we need dataset. So, during the first step of KNN, we must load the training as well as test data.

Step 2 − Next, we need to choose the value of K i.e. the nearest data points. K can be any integer.

Step 3 − For each point in the test data do the following

3.1 − Calculate the distance between test data and each row of training data with the help of any of the method namely: Euclidean, Manhattan or Hamming distance. The most commonly used method to calculate distance is Euclidean.

3.2 − Now, based on the distance value, sort them in ascending order.

3.3 − Next, it will choose the top K rows from the sorted array.

3.4 − Now, it will assign a class to the test point based on most frequent class of these rows.

Step 4 – End

Using Euclidian distance

Example : Input

- Based on two attributes : Acid durability and strength, we want to classify a paper tissue into good/bad quality classes :

Independent variables/predictors

Target Variable (Y)

Acid durability (in Seconds)	Strength (Kg/Square meter)	Paper tissue Quality
7	7	Good
7	4	Good
3	4	Bad
1	4	Good

Based on this classification:

- If a **paper tissue** has **Acid Durability = 3** and **Strength = 7**, **what will be its quality?**

Example : Steps :

Calculate Distance and Ranking to find K nearest neighbors

- For instance :
  - If a paper tissue's acid durability = 3 and strength = 7 then take following steps :
    - Step 1 : Decide value of k ; Say it is 3 (based on classification accuracy)

Acid durability (In Seconds)	Strength (Kg/Square meter)	Paper tissue Quality	Distance to instance	Rank by distance to find nearest neighbor
7	7	Good	$(7-3)^2 + (7-7)^2 = 16$	3
7	4	Good	$(7-3)^2 + (4-7)^2 = 25$	4
3	4	Bad	$(3-3)^2 + (4-7)^2 = 9$	1
1	4	Good	$(1-3)^2 + (4-7)^2 = 13$	2

Input dataset

Derived results to find out the majority class of k nearest neighbors

Example : Steps

Select majority class of k nearest neighbors as predicted class

Step 4,5 : As the majority class = Good for the three nearest neighbors (two out of three records have class = Good) , predicted class of an instance = Good, i.e. quality of a paper tissue having acid durability =3 and strength =7 is good

Final output :

Acid durability (In Seconds)	Strength (Kg/Square meter)	Paper tissue Quality
7	7	Good
7	4	Good
3	4	Bad
1	4	Good
3	7	Good

Example : Steps
Find out Accuracy

Predicted

Actual

	Good	Bad
Good	35	4
Bad	4	70

CLASSIFICATION ACCURACY : *(35+ 70) / (35+70+4+4) = 92%*

- *The prediction accuracy is useful criterion for assessing the model performance*
- *Model with prediction accuracy >= 70% is useful*

CLASSIFICATION ERROR *= 100- Accuracy = 8%*

There is 8% chance of error in classification

Using Manhattan distance

Acid Durability	Strength	Quality
4	6	Medium
4	5	Medium
5	8	High
6	3	Low
3	7	Medium
8	4	Medium
7	9	High
1	2	Low
9	10	High
10	1	Low

Based on this classification: **K=3**

- If a **paper tissue** has **Acid Durability = 5** and **Strength = 5**, **what will be its quality using Manhattan distance.**

Acid Durability(A)	Strength(S)	Quality	Manhattan |A-5|+|S-5|	RESULT
4	6	Medium	1+1	2
4	5	Medium	1+0	1
5	8	High	0+3	3
6	3	Low	1+2	3
3	7	Medium	2+2	4
8	4	Medium	3+1	4
7	9	High	2+4	6
1	2	Low	4+3	7
9	10	High	4+5	9
10	1	Low	5+4	9

Based on this classification: **K=3 Majority class : Medium**
If a **paper tissue** has **Acid Durability = 5** and **Strength = 5**, **its quality is Medium**

Using Cosine similarity

Point	Feature 1	Feature 2	Class
A	1	2	0
B	2	3	0
C	4	3	1
D	5	6	1

a new test point **(Feature 1 = 4, Feature 2 = 5)**, and we need to classify it using **KNN (K = 3)** with **Cosine Similarity**.

Formula:

the cosine similarity between two vectors A and B is given by:

$$\text{cosine similarity} = \frac{A \cdot B}{||A|| \times ||B||}$$

Step 1: Compute Dot Products

For each training point $X = (x_1, x_2)$:

$$\text{dot product} = (4 \times x_1) + (5 \times x_2)$$

- Test vs A: $(4 \times 1) + (5 \times 2) = 4 + 10 = 14$
- Test vs B: $(4 \times 2) + (5 \times 3) = 8 + 15 = 23$
- Test vs C: $(4 \times 4) + (5 \times 3) = 16 + 15 = 31$
- Test vs D: $(4 \times 5) + (5 \times 6) = 20 + 30 = 50$

Step 2: Compute Norms

$$||A|| = \sqrt{a_1^2 + a_2^2}$$

For each point:

- Test point: $||(4,5)|| = \sqrt{4^2 + 5^2} = \sqrt{41} = 6.4$
- A (1,2): $||(1,2)|| = \sqrt{1^2 + 2^2} = \sqrt{5} = 2.24$
- B (2,3): $||(2,3)|| = \sqrt{2^2 + 3^2} = \sqrt{13} = 3.61$
- C (4,3): $||(4,3)|| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.0$
- D (5,6): $||(5,6)|| = \sqrt{5^2 + 6^2} = \sqrt{61} = 7.81$

Step 3: Compute Cosine Similarities

$$\cos(\theta) = \frac{\text{dot product}}{||A|| \times ||B||}$$

- Test vs A: $\frac{14}{(6.4 \times 2.24)} = 0.98$
- Test vs B: $\frac{23}{(6.4 \times 3.61)} = 0.996$
- Test vs C: $\frac{31}{(6.4 \times 5.0)} = 0.968$
- Test vs D: $\frac{50}{(6.4 \times 7.81)} = 1.00$

Point	Feature 1	Feature 2	Class	Cosine similarity	Rank
A	1	2	0	0.98	2
B	2	3	0	0.99	3
C	4	3	1	0.96	1
D	5	6	1	1.0	4

Majority class: **0**
Thus, the new test point **(4,5) is classified as Class 0**.

Example of weighted k-NN classification:

Feature 1	Feature 2	Target Class
1.2	3.5	B
2.3	4.2	A
1.9	1.8	B
4.5	5.0	B
4.5	1.2	A
5.5	3.8	B
2.6	3.2	A

use weighted k-NN to predict dataset with the following features and target classes:

Feature 1	Feature 2	Target Class
3.0	3.5	?

Decide the value of k = 3(based on classification accuracy) and where p = 2.

weight = 1 / distance^p

Calculate the Euclidean distance between the new data point and each point in the training set.

Feature 1	Feature 2	Target Class	Distance
1.2	3.5	B	1.8
2.3	4.2	A	1.3
1.9	1.8	B	2.0
4.5	5.0	B	2.8
4.5	1.2	A	2.5
5.5	3.8	B	2.7
2.6	3.2	A	0.5

2.Select k nearest neighbors: Select the k nearest neighbors to the new data point based on their distances. In this case, k = 3, so we select the three points with the smallest distances:

Feature 1	Feature 2	Target Class	Distance
2.6	3.2	A	0.5
2.3	4.2	A	1.3
1.2	3.5	B	1.8

3.Calculate weights: Calculate the inverse distance weights for the k nearest neighbors using the formula **weight = 1 / d^p** where d is the distance to the i-th nearest neighbor and p is a tuning parameter. In this case, p = 2.

Feature 1	Feature 2	Target Class	Distance	Weight
2.6	3.2	A	0.5	4.0
2.3	4.2	A	1.3	0.59
1.2	3.5	B	1.8	0.30

Since the weighted votes for class A are higher than class B, the test instance is classified as class A.

Customer	Age	Income	Buys Product? (Class)
A	25	50000	0
B	35	60000	1
C	45	80000	1
D	20	30000	0
E	40	70000	1

classify a **new customer (Test Point: Age = 38, Income = 65,000)** using **Weighted KNN Classification (K=3)**.

Difference between Regression and Classification

Regression Algorithm	**Classification Algorithm**
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier

Performance metrics

Mean Absolute Error(MAE):-	MAE is a very simple metric which calculates the absolute difference between actual and predicted values.	$MAE = \frac{1}{n}\sum_{i=1}^{N}(Predicted_i - Actual_i)$
Mean Squared Error(MSE):-	Mean squared error states that finding the squared difference between actual and predicted value	$MSE = \frac{1}{n}\sum_{i=1}^{N}(Predicted_i - Actual_i)^2$
Root Mean Squared Error(RMSE):-	As RMSE is clear by the name itself, that it is a simple square root of mean squared error	$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(Predicted_i - Actual_i)^2}{N}}$
Mean Absolute Percentage Error(MAPE)	MAPE which is a commonly used metric for evaluating the accuracy of a forecasting model.	$MAPE = \frac{1}{n} \times \sum \left\lvert \frac{actual\ value - forecast\ value}{actual\ value} \right\rvert$

KNN for Regression [5]

- KNN for Regression

Age	Loan	House Price Index	Distance	
25	$40,000	135	102000	
35	$60,000	256	82000	
45	$80,000	231	62000	
20	$20,000	267	122000	
35	$120,000	139	22000	2
52	$18,000	150	124000	
23	$95,000	127	47000	
40	$62,000	216	80000	
60	$100,000	139	42000	3
48	$220,000	250	78000	
33	$150,000	264	8000	1
48	**$142,000**	**?**		

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

KNN for Regression [5]

- We can now use the training set to classify an unknown case (Age=48 and Loan=$142,000) using Euclidean distance.

- If K=1 then the nearest neighbour is the last case in the training set with HPI=264.

- D = Sqrt[(48-33)^2 + (142000-150000)^2] = 8000.01 >> HPI = 264

- By having K=3, the prediction for HPI is equal to the average of HPI for the top three neighbors.

- HPI = (264+139+139)/3 = 180.7

KNN for Regression [5]

- Important thing to notice in the given data is to notice the difference in the scale of Age and Loan.
- We should ideally normalize the data in such cases.

Example of k-NN regression:

Suppose we have a dataset of housing prices and we want to use 3-NN regression to predict the price of a new house. The dataset has the following features: square footage, number of bedrooms, number of bathrooms, and price.

Square Footage	Bedrooms	Bathrooms	Price
1000	2	1	150
1500	3	2	250
2000	4	3	400
2500	4	2	500
3000	5	3	600

Predict the price of a new house with the following features: 1800 square footage, 3 bedrooms, and 2 bathrooms.

Square Footage	Bedrooms	Bathrooms	Price
1800	3	2	?

1.Calculate the distances: We need to calculate the Euclidean distance between the new house and each house in the dataset. The distances are:

Square Footage	Bedrooms	Bathrooms	Price	Distance
1000	2	1	150	800
1500	3	2	250	300
2000	4	3	400	200
2500	4	2	500	700
3000	5	3	600	1200

2.Choose the k-nearest neighbors: We select the k-nearest neighbors from the distances we calculated. In this example, we choose k = 3, so the three nearest neighbors are:

Square Footage	Bedrooms	Bathrooms	Price
1500	3	2	250
2000	4	3	400
2500	4	2	500

3.Calculate the predicted price: We use the average of the prices of the k-nearest neighbors as the predicted price of the new house. In this example, the predicted price is:

Predicted price = (250 + 400 + 500) / 3 = 383.33

Therefore, the k-NN regression algorithm predicts that the price of the new house with 1800 square footage, 3 bedrooms, and 2 bathrooms is:

Square Footage	Bedrooms	Bathrooms	Price
1800	3	2	383.33

Predict the mileage of a car with an **engine size of 1.6L and weight of 1350 kg**, using **KNN Regression (K=3) with Manhattan Distance**.

Engine Size (L)	Weight (kg)	Mileage (mpg)
1.2	1100	45
1.5	1300	40
1.8	1400	35
2	1600	30
2.2	1800	25
1.6	1350	?

KNN for Regression [5]

- Important thing to notice in the given data is to notice the difference in the scale of Age and Loan.
- We should ideally normalize the data in such cases.

If the features are on **different scales**, larger numerical values will dominate the distance calculations, leading to **biased or incorrect predictions**.

Min-Max Scaling Formula

Min-Max scaling transforms data to a fixed range [0,1]:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

KNN for Regression [5]

- KNN for Regression

Age	Loan	House Price Index	Distance	
25	$40,000	135	102000	
35	$60,000	256	82000	
45	$80,000	231	62000	
20	$20,000	267	122000	
35	$120,000	139	22000	2
52	$18,000	150	124000	
23	$95,000	127	47000	
40	$62,000	216	80000	
60	$100,000	139	42000	3
48	$220,000	250	78000	
33	$150,000	264	8000	1
48	**$142,000**	**?**		

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Step1: Compute Min & Max for Each Feature

Feature	Min Value	Max Value
Age	20	60
Loan ($)	18,000	2,20,000

Step 3: Apply Min-Max Scaling Formula

$$X_s = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Original Age	Original Loan ($)	Scaled Age	Scaled Loan
25	40,000	$\frac{25-20}{60-20} = \frac{5}{40} = 0.125$	$\frac{40{,}000-18{,}000}{220{,}000-18{,}000} = \frac{22{,}000}{202{,}000} = 0.1089$
35	60,000	$\frac{35-20}{60-20} = \frac{15}{40} = 0.375$	$\frac{60{,}000-18{,}000}{220{,}000-18{,}000} = \frac{42{,}000}{202{,}000} = 0.2079$
45	80,000	$\frac{45-20}{60-20} = \frac{25}{40} = 0.625$	$\frac{80{,}000-18{,}000}{220{,}000-18{,}000} = \frac{62{,}000}{202{,}000} = 0.3079$
20	20,000	$\frac{20-20}{60-20} = \frac{0}{40} = 0.000$	$\frac{20{,}000-18{,}000}{220{,}000-18{,}000} = \frac{2{,}000}{202{,}000} = 0.0099$
35	120,000	$\frac{35-20}{60-20} = \frac{15}{40} = 0.375$	$\frac{120{,}000-18{,}000}{220{,}000-18{,}000} = \frac{102{,}000}{202{,}000} = 0.5049$
52	18,000	$\frac{52-20}{60-20} = \frac{32}{40} = 0.800$	$\frac{18{,}000-18{,}000}{220{,}000-18{,}000} = \frac{0}{202{,}000} = 0.0000$
23	95,000	$\frac{23-20}{60-20} = \frac{3}{40} = 0.075$	$\frac{95{,}000-18{,}000}{220{,}000-18{,}000} = \frac{77{,}000}{202{,}000} = 0.3812$
40	62,000	$\frac{40-20}{60-20} = \frac{20}{40} = 0.500$	$\frac{62{,}000-18{,}000}{220{,}000-18{,}000} = \frac{44{,}000}{202{,}000} = 0.2178$
60	100,000	$\frac{60-20}{60-20} = \frac{40}{40} = 1.000$	$\frac{100{,}000-18{,}000}{220{,}000-18{,}000} = \frac{82{,}000}{202{,}000} = 0.4059$
48	220,000	$\frac{48-20}{60-20} = \frac{28}{40} = 0.700$	$\frac{220{,}000-18{,}000}{220{,}000-18{,}000} = \frac{202{,}000}{202{,}000} = 1.000$
33	150,000	$\frac{33-20}{60-20} = \frac{13}{40} = 0.325$	$\frac{150{,}000-18{,}000}{220{,}000-18{,}000} = \frac{132{,}000}{202{,}000} = 0.6535$

Step 4: Scaled Data

KNN for Regression [5]

Age	Loan	House Price Index	Distance
0.125	0.11	135	0.7652
0.375	0.21	256	0.5200
0.625	0.31	231	0.3160
0	0.01	267	0.9245
0.375	0.50	139	0.3428
0.8	0.00	150	0.6220
0.075	0.38	127	0.6669
0.5	0.22	216	0.4437
1	0.41	139	0.3650
0.7	1.00	250	0.3861
0.325	0.65	264	0.3771
0.7	**0.61**	**?**	

$$X_s = \frac{X - Min}{Max - Min}$$

Predict the wheat crop yield for **1100mm rainfall and 77% soil fertility**, using **KNN Regression (K=3) with Min-Max feature scaling** Using **Manhattan distance**

Rainfall (mm)	Soil Fertility (%)	Yield (tons/ha)
800	70	3
1000	75	3.5
1200	80	4.2
1500	85	5
1800	90	5.8
1100	77	?

Weighted KNN Regression

Given the following dataset, predict the **price of a house with Area = 2300 sqft and Bedrooms = 3** using **Weighted KNN Regression** with **k=3**.

Area (sqft)	Bedrooms	Price ($1000s)
1800	2	150
2200	3	180
3000	4	250
2500	2	210
2700	3	230
2600	3	175
2300	**3**	**?**

Steps:

1.Compute Distance: Find the distance between the query point and all dataset points.

2.Select k Nearest Neighbors: Pick the three closest points.

3.Compute Weights: Use inverse distance weighting.

$$w_i = \frac{1}{d_i}$$

4. Compute Weighted Average: Use the formula:

$$\hat{y} = \frac{\sum (w_i \times y_i)}{\sum w_i}$$

5. Final Prediction

Step 1: Compute Euclidean Distance

For each point in the dataset, compute the **Euclidean distance** to the query point (x_q, y_q):

$$d = \sqrt{(x - x_q)^2 + (y - y_q)^2}$$

Area (sqft)	Bedrooms	Price ($1000s)	Distance
1800	2	150	500
2200	3	180	100
3000	4	250	700
2500	2	210	200
2700	3	230	400
2600	3	175	300
2300	**3**	**?**	

Step 2: Find the 3 Nearest Neighbors

We **sort the distances** in ascending order and pick the top **3 nearest neighbors**:

Area (sqft)	Bedrooms	Price ($1000s)	Distance
2200	3	180	100
2500	2	210	200
2600	3	175	300

Step 3: Compute Weights

The weight for each neighbor is calculated using the **inverse of the distance**:

$$w_i = \frac{1}{d_i}$$

Area (sqft)	Bedrooms	Price ($1000s)	Distance	Weight(w=1/d)
2200	3	180	100	.01
2500	2	210	200	.005
2600	3	175	300	.003

Step 4: Compute Weighted Prediction

The final price prediction is calculated using the **weighted average formula**:

$$\hat{y} = \frac{\sum(w_i \times y_i)}{\sum w_i}$$

Substituting values:

$$\hat{y} = \frac{(0.01000 \times 180) + (0.00500 \times 210) + (0.00333 \times 175)}{0.01000 + 0.00500 + 0.00333}$$

$$\hat{y} = \frac{1.8 + 1.05 + 0.583}{0.01833}$$

$$\hat{y} = \frac{3.433}{0.01833} = 187.27$$

Final Answer:
The **predicted house price** for (2300 sqft, 3 bedrooms) using **Weighted KNN Regression with k=3** is:187

Area (sqft)	Bedrooms	Price ($1000s)
2300	**3**	**187**

Example:

Prediction Target:

- Car with Engine Size = 1700cc, Mileage = 20 km/l
- Apply Weighted KNN Regression (k = 3)
- Using Manhattan distance
- Find weighted regression

Engine Size (cc)	Mileage (km/l)	Car Price ($1000s)
1200	18	15
1500	22	20
1800	16	28
2000	20	35
1600	19	24
1400	21	18
1700	20	?

Step 1:

Engine Size (cc)	Mileage (km/l)	Car Price ($1000s)	Distance
1200	18	15	502
1500	22	20	202
1800	16	28	104
2000	20	35	300
1600	19	24	101
1400	21	18	321

Step 2:

Engine Size (cc)	Mileage (km/l)	Car Price ($1000s)	Distance	Weight
1600	19	24	101	0.001
1800	16	28	104	0.001
1500	22	20	202	0.005

3. **Apply Weighted KNN Formula**

$$\hat{y} = \frac{\sum(w_i \times y_i)}{\sum w_i}$$

$$\hat{y} = \frac{(0.0099 \times 24) + (0.0096 \times 28) + (0.0050 \times 20)}{0.0099 + 0.0096 + 0.0050}$$

$$\hat{y} = 24.76K$$

Answer:

Engine Size (cc)	Mileage (km/l)	Car Price ($1000s)
1700	**20**	**24**

K-Means Algorithm

K-Means Clustering is an **Unsupervised Learning algorithm**, It is an iterative algorithm that divides the unlabeled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.

Clustering :

- It is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.
- It is basically a collection of objects on the basis of similarity and dissimilarity between them.

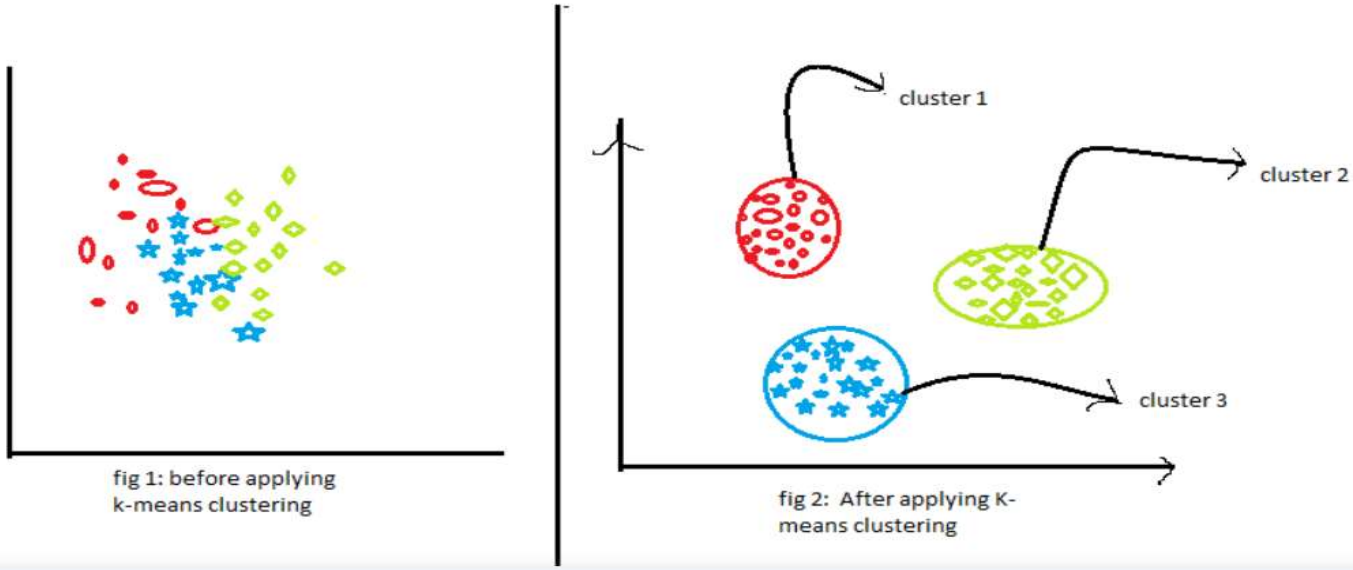


fig 1: before applying k-means clustering

fig 2: After applying K-means clustering

Unsupervised Learning

- K-means Algorithm [1]

Algorithm: *k*-means. The *k*-means algorithm for partitioning, where each cluster's center is represented by the mean value of the objects in the cluster.

Input:

- *k*: the number of clusters,
- *D*: a data set containing *n* objects.

Output: A set of *k* clusters.

Method:

(1) arbitrarily choose *k* objects from *D* as the initial cluster centers;
(2) **repeat**
(3) (re)assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster;
(4) update the cluster means, i.e., calculate the mean value of the objects for each cluster;
(5) **until** no change;

K-Means Clustering Algorithm-

K-Means Clustering Algorithm involves the following steps-

Step-01:

- Choose the number of clusters K.

Step-02:

- Randomly select any K data points as cluster centers.
- Select cluster centers in such a way that they are as farther as possible from each other.

Step-03:

- Calculate the distance between each data point and each cluster center.
- The distance may be calculated either by using given distance function or by using euclidean distance formula.

Step-04:

- Assign each data point to some cluster.
- A data point is assigned to that cluster whose center is nearest to that data point.

Step-05:

- Re-compute the center of newly formed clusters.
- The center of a cluster is computed by taking mean of all the data points contained in that cluster.

Step-06:

- Keep repeating the procedure from Step-03 to Step-05 until any of the following stopping criteria.
- Center of newly formed clusters do not change
- Data points remain present in the same cluster
- Maximum number of iterations are reached

PROBLEMS BASED ON K-MEANS CLUSTERING ALGORITHM-

Cluster the following eight points (with (x, y) representing locations) into three clusters:
A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).
The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as-

$$P(a, b) = |x2 – x1| + |y2 – y1|$$

Use K-Means Algorithm to find the three cluster centers after the **second iteration.**

Solution-

We follow the above discussed K-Means Clustering Algorithm-

Iteration-01:

•We calculate the distance of each point from each of the center of the three clusters.
•The distance is calculated by using the given distance function.

The following illustration shows the calculation of distance between point A1(2, 10) and each of the center of the three clusters-

Calculating Distance Between A1(2, 10) and C1(2, 10)-

P(A1, C1)
= |x2 – x1| + |y2 – y1|
= |2 – 2| + |10 – 10|
= 0

Calculating Distance Between A1(2, 10) and C2(5, 8)-

P(A1, C2)
= |x2 – x1| + |y2 – y1|
= |5 – 2| + |8 – 10|
= 3 + 2
= 5

Calculating Distance Between A1(2, 10) and C3(1, 2)-

P(A1, C3)
= |x2 – x1| + |y2 – y1|
= |1 – 2| + |2 – 10|
= 1 + 8
= 9

In the similar manner, we calculate the distance of other points from each of the center of the three clusters.

Next,
•We draw a table showing all the results.
•Using the table, we decide which point belongs to which cluster.
•The given point belongs to that cluster whose center is nearest to it.

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (5, 8) of Cluster-02	Distance from center (1, 2) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	5	9	C1
A2(2, 5)	5	6	4	C3
A3(8, 4)	12	7	9	C2
A4(5, 8)	5	0	10	C2
A5(7, 5)	10	5	9	C2
A6(6, 4)	10	5	7	C2
A7(1, 2)	9	10	0	C3
A8(4, 9)	3	2	10	C2

From here, New clusters are-

Cluster-01:
First cluster contains points-
•A1(2, 10)

Cluster-02:
Second cluster contains points-
•A3(8, 4)
•A4(5, 8)
•A5(7, 5)
•A6(6, 4)
•A8(4, 9)

Cluster-03:
Third cluster contains points-
•A2(2, 5)
•A7(1, 2)

Now,
•We re-compute the new cluster clusters.
•The new cluster center is computed by taking mean of all the points contained in that cluster.

For Cluster-01:

•We have only one point A1(2, 10) in Cluster-01.
•So, cluster center remains the same.

For Cluster-02:

Center of Cluster-02
= ((8 + 5 + 7 + 6 + 4)/5, (4 + 8 + 5 + 4 + 9)/5)
= (6, 6)

For Cluster-03:

Center of Cluster-03
= ((2 + 1)/2, (5 + 2)/2)
= (1.5, 3.5)

This is completion of Iteration-01.

<u>Iteration-02:</u>

•We calculate the distance of each point from each of the center of the three clusters.
•The distance is calculated by using the given distance function.

The following illustration shows the calculation of distance between point A1(2, 10) and each of the center of the three clusters-

Calculating Distance Between A1(2, 10) and C1(2, 10)-

P(A1, C1)
= |x2 – x1| + |y2 – y1|
= |2 – 2| + |10 – 10|
= 0

Calculating Distance Between A1(2, 10) and C2(6, 6)-

P(A1, C2)
= |x2 – x1| + |y2 – y1|
= |6 – 2| + |6 – 10|
= 4 + 4
= 8

Calculating Distance Between A1(2, 10) and C3(1.5, 3.5)-

P(A1, C3)
= |x2 – x1| + |y2 – y1|
= |1.5 – 2| + |3.5 – 10|
= 0.5 + 6.5
= 7

In the similar manner, we calculate the distance of other points from each of the center of the three clusters.

Next,

•We draw a table showing all the results.

•Using the table, we decide which point belongs to which cluster.

•The given point belongs to that cluster whose center is nearest to it.

Given Points	Distance from center (2, 10) of Cluster-01	Distance from center (6, 6) of Cluster-02	Distance from center (1.5, 3.5) of Cluster-03	Point belongs to Cluster
A1(2, 10)	0	8	7	C1
A2(2, 5)	5	5	2	C3
A3(8, 4)	12	4	7	C2
A4(5, 8)	5	3	8	C2
A5(7, 5)	10	2	7	C2
A6(6, 4)	10	2	5	C2
A7(1, 2)	9	9	2	C3
A8(4, 9)	3	5	8	C1

From here, New clusters are-

Cluster-01:
First cluster contains points-
•A1(2, 10)
•A8(4, 9)

Cluster-02:
Second cluster contains points-
•A3(8, 4)
•A4(5, 8)
•A5(7, 5)
•A6(6, 4)

Cluster-03:
Third cluster contains points-
Third cluster contains points-
•A2(2, 5)
•A7(1, 2)

Now,
•We re-compute the new cluster clusters.
•The new cluster center is computed by taking mean of all the points contained in that cluster.

For Cluster-01:

Center of Cluster-01
= ((2 + 4)/2, (10 + 9)/2)
= (3, 9.5)

For Cluster-02:

Center of Cluster-02
= ((8 + 5 + 7 + 6)/4, (4 + 8 + 5 + 4)/4)
= (6.5, 5.25)

For Cluster-03:

Center of Cluster-03
= ((2 + 1)/2, (5 + 2)/2)
= (1.5, 3.5)

This is completion of Iteration-02.

After second iteration, the center of the three clusters are-
- C1(3, 9.5)
- C2(6.5, 5.25)
- C3(1.5, 3.5)